

PROJECT TITLE: EHR Companion for Health Workers

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ABSTRACT

Healthcare delivery in rural India largely depends on Accredited Social Health Activist (ASHA) workers and Primary Health Centre (PHC) staff. However, maintaining accurate Electronic Health Records (EHRs) is a major challenge due to low or inconsistent internet connectivity in these regions. Existing digital health solutions often fail in such contexts, leading to delays in patient record updates, gaps in vaccination tracking, and reduced efficiency of national health programs.

This project proposes the development of a **Web-based EHR Companion** designed using **full-stack web technologies** and enhanced with **Generative AI (GenAI)** capabilities. The system will enable healthcare workers to collect and store patient information offline during field visits, with **automatic synchronization** once internet connectivity is available. It will feature a **simple and intuitive interface**, **regional language support**, and **voice-based input** to ensure accessibility for semi-literate users.

Role-based dashboards will provide tailored views for ASHA and PHC staff, offering actionable insights such as **vaccination reminders**, **antenatal care (ANC) schedules**, and follow-up alerts. Additionally, the integration of GenAI will allow the system to **generate patient history summaries**, **assist in clinical decision-making**, and deliver **personalized recommendations**, thereby reducing cognitive load on health workers.

The proposed EHR Companion addresses the pressing challenge of digital healthcare access in low-internet areas. By combining **offline-first design**, **full-stack web development**, and **AI-driven assistance**, it empowers health workers with reliable tools to manage patient data effectively. This solution not only improves healthcare service delivery in underserved communities but also supports the broader objectives of national health initiatives, bridging the digital divide in India's healthcare ecosystem.

